3. Write the nth term of the following sequences.

(i) $2,\ 2,\ 4,\ 4,\ 6,\ 6,\ \ldots$ (ii) $\frac{1}{2},\ \frac{2}{3},\ \frac{3}{4},\ \frac{4}{5},\ \frac{5}{6},\ \ldots$
(iii) $\frac{1}{2},\ \frac{3}{4},\ \frac{5}{6},\ \frac{7}{8},\ \frac{9}{10},\ \ldots$ (iv) $6,\ 10,\ 4,\ 12,\ 2,\ 14,\ 0, 16,\ -2, \ldots$

4. The product of three increasing numbers in GP is 5832. If we add 6 to the second number and 9 to the third number, then resulting numbers form an AP. Find the numbers in GP.
5. Write the nth term of the sequence $\frac{3}{1^2 2^2}, \frac{5}{2^2 3^2}, \frac{7}{3^2 4^2}, \ldots$ as a difference of two terms.
6. If t_k is the kth term of a GP, then show that t_{n-k}, t_n, t_{n+k} also form a GP for any positive integer k.
7. If a, b, c are in geometric progression, and if $a^{\frac{1}{x}} = b^{\frac{1}{y}} = c^{\frac{1}{z}}$, then prove that x, y, z are in arithmetic progression.
8. The AM of two numbers exceeds their GM by 10 and HM by 16. Find the numbers.
9. If the roots of the equation $(q-r)x^2 + (r-p)x + p - q = 0$ are equal, then show that p, q and r are in AP.
10. If a, b, c are respectively the pth, q^{th} and r^{th} terms of a GP, show that $(q-r)\log a + (r-p)\log b + (p-q)\log c = 0$.

5.5 Finite Series



Roughly speaking a series is the sum of terms of a sequence of numbers; a finite series is the sum of terms of a finite sequence of numbers. If (a_n) is a sequence of numbers, then the expression $a_1 + a_2 + \cdots + a_n$ is called a ***finite series***. The expression $a_1 + a_2 + \cdots + a_n$ is denoted as $\sum_{k=1}^{n} a_k$. Sometimes, depending upon the problem under consideration and for simplicity a series may be given as $a_0 + a_1 + a_2 + \cdots$ with first term as a_0.

5.5.1 Sum of Arithmetic, Geometric and Arithmetico-Geometric Progressions

In the earlier classes we studied about the sum of a few terms, like sum of first n terms, of arithmetic and geometric progressions. We now recall them.

Sum of Arithmetic and Geometric Progressions

- A series is said to be an ***arithmetic series*** if the terms of the series form an arithmetic sequence. A series is said to be a ***geometric series*** if the terms of the series form a geometric sequence.
- The sum S_n of the first n terms of the arithmetic sequence $(a + (n-1)d)$ is given by $S_n = na + \frac{(n-1)n}{2}d = \frac{n}{2}[2a + (n-1)d]$.
- The sum S_n of the first n terms of the geometric sequence (ar^{n-1}) is given by $S_n = \frac{a(1-r^n)}{1-r}$ provided $r \neq 1$. If $r = 1$, then the sequence is nothing but the constant sequence $a, a, a, \ldots$ and the sum of the first n terms is clearly na. Thus, if $r \neq 1$, then $1 + r + r^2 + \cdots + r^{n-1} = \frac{1-r^n}{1-r}$.

Sum of Arithmetico-Geometric Progressions

- A series is said to be an ***arithmetico-Geometric series*** if the terms of the series form an arithmetico-Geometric sequence.
- The sum S_n of the first n terms of the arithmetico-Geometric sequence $((a + (n-1)d)r^{n-1})$ is given by

$$S_n = \frac{a - (a + (n-1)d)r^n}{1 - r} + dr\left(\frac{1 - r^{n-1}}{(1-r)^2}\right)$$

for $r \neq 1$.